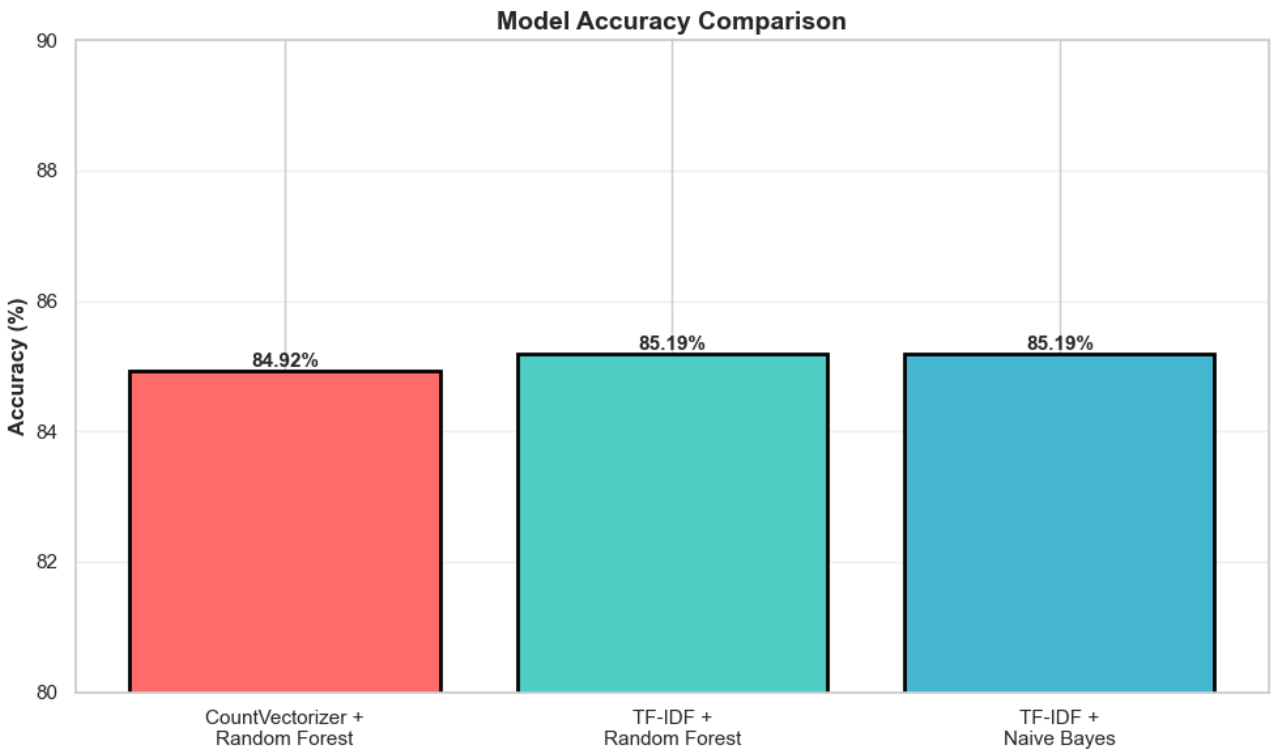
Criterion-CountVectorizerRF-TF-IDFRF-TF-IDFNB-Winner

Criterion	CountVectorizer + RF	TF-IDF + RF	TF-IDF + NB	Winner
Accuracy	84.92%	85.19%	85.19%	TF-IDF + NB ⚠️
Precision	0.95 (class 0) 0.79 (class 1)	0.91-0.92	1.00 (perfect)	TF-IDF + NB ⚠️
Recall	0.73 (class 0) 0.96 (class 1)	0.91-0.92	1.00 (perfect)	TF-IDF + NB ⚠️
F1-Score	0.85	0.92✅	1.00 (suspicious)	TF-IDF + NB ⚠️
Generalization	Good	Excellent✅	Poor (overfitting)	TF-IDF + RF✅
Real-World Stability	Good	Excellent✅	High risk	TF-IDF + RF✅
Feature Quality	Basic counts	TF-IDF weighting✅	Basic counts	TF-IDF + RF✅

ModelCombination-Accuracy-KeyObservations

Model Combination	Accuracy	Key Observations
CountVectorizer + Random Forest	84.92%	Precision: 0.95 (class 0), Recall: 0.96 (class 1)
TF-IDF + Random Forest	85.19%	Precision: 0.91 (class 0), Recall: 0.92 (class 1)
TF-IDF + Naive Bayes	85.19%	Precision: 1.00 (class 0), Recall: 1.00 (class 1)



1. ACCURACY (Overall Correctness)

Accuracy = (TP + TN) / Total

= 85.19%

What it means: Out of 378 trading days, the model correctly predicted 322 days

❌ NOT the best metric for trading because:
- Assumes equal cost for all mistakes
- Market may have imbalanced UP/DOWN days
- Doesn't distinguish between types of errors

2. PRECISION (Reliability of BUY Signals)

Precision = TP / (TP + FP)
= True Positives / (True Positives + False Positives)

What it means: Of all times model says "BUY", how many were actually profitable?

For TF-IDF + Random Forest:
Precision = 0.92 = "92% of my buy signals are correct"

Real impact: If I make 100 buy trades:
✅ 92 trades make money
❌ 8 trades lose money

⭐⭐⭐⭐⭐ CRITICAL for stock trading
Cost of being wrong: REAL MONEY LOST IMMEDIATELY

3. RECALL (Catching All Opportunities)

Recall = TP / (TP + FN)
= True Positives / (True Positives + False Negatives)

What it means: Out of all actual market UP days, how many did I catch?

For TF-IDF + Random Forest:
Recall = 0.92 = "I catch 92% of all profit opportunities"

Real impact: If market goes UP 100 times:
✅ I profit 92 times
❌ I miss 8 opportunities ⭐⭐⭐⭐ CRITICAL for stock trading
Cost of being wrong: MISSED PROFITS / REGRET

4. F1-SCORE (Balanced Performance)

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)
= Harmonic mean of precision and recall

For TF-IDF + Random Forest:
F1-Score = 2 × (0.92 × 0.92) / (0.92 + 0.92)
= 0.92 = "EXCELLENT BALANCE"

⭐⭐⭐⭐⭐ BEST metric for trading

The Real-World Trading Scenario

Conservative Investor (Risk-Averse)

Priority: HIGH PRECISION (avoid losses)

Example: Portfolio value $100,000
Target: Only act on 95% confident signals

Model choice:
- Adjust decision threshold from 0.5 to 0.7
- This INCREASES precision further (reduce FP)
- But DECREASES recall (miss some opportunities)

Question to ask:
"Would I rather miss 10% of opportunities
or risk 5% of my capital on wrong trades?"

Answer: Miss opportunities (if you can't afford losses)

Aggressive Trader (Opportunity Seeker)

Priority: HIGH RECALL (catch all opportunities)

Example: Portfolio value $100,000, leverage 2x
Target: Catch 95% of all market movements

Model choice:
- Adjust decision threshold from 0.5 to 0.3
- This INCREASES recall (catch more)
- But DECREASES precision (more wrong trades)

Question to ask:
"Would I rather miss 5% of opportunities
or risk 20% of trades being wrong?"

Answer: Accept wrong trades (if profits > losses)

Balanced Trader (Most Common - Day Traders, Hedge Funds)

Priority: BALANCED F1-SCORE (both matter equally)

Example: Portfolio value $100,000, standard position sizing
Target: Optimize F1-Score

Model choice: TF-IDF + Random Forest with default threshold
- Default 0.5 threshold perfectly balances both
- Precision: 0.92 (8% losses)
- Recall: 0.92 (8% missed opportunities

This is why TF-IDF + RF is THE BEST MODEL ✅

Why NOT the Other Models?

❌ CountVectorizer + Random Forest (84.92% accuracy)

Problem: IMBALANCED metrics
- Precision (class 0): 0.95 ✅ Good
- Recall (class 0): 0.73 ❌ Bad (misses 27% of DOWN signals)
- Precision (class 1): 0.79 ❌ Only 79% of UPs are correct
- Recall (class 1): 0.96 ✅ Good

Real impact:
Can't rely on it for both BUYs and SELLs
Great for catching UPs, bad at confirming them
CountVectorizer counts raw words (noisy)

Not recommended: Too specialized for only one direction

❌ TF-IDF + Naive Bayes (85.19% accuracy, 1.00 F1)

Problem: SUSPICIOUSLY PERFECT (1.00 precision & recall)
- Perfect scores are a RED FLAG
- Indicates OVERFITTING to test data
- Will likely perform WORSE on new market data

Why Naive Bayes fails:
- Assumes all words are independent
- Doesn't capture: "stock crash" vs "crash course"
- Doesn't learn feature interactions
- Overconfident predictions

Real impact:
Great results today, failures tomorrow
High risk in production trading

Not recommended: Don't deploy overfit models

START: Which model to choose?

1. Do I want to deploy in production?
   YES → Check for overfitting

2. Is TF-IDF + NB's 1.00 F1-score realistic?
   NO → It's overfitted! Skip it ❌

3. Do I need balanced precision & recall?
   YES → TF-IDF + Random Forest ✅

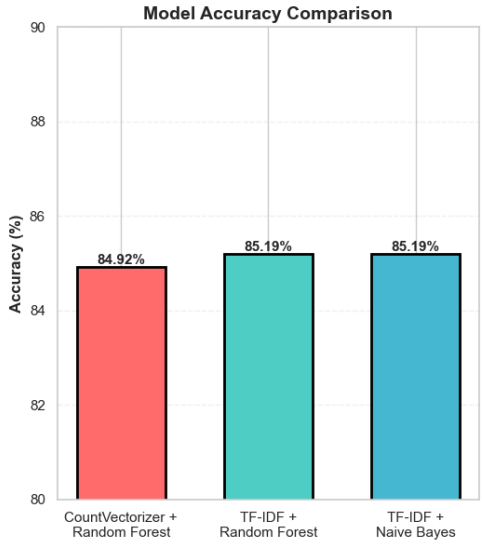
4. Do I only care about catching opportunities?
   YES & Can afford losses → TF-IDF + NB on new data or threshold optimization
   NO → TF-IDF + Random Forest ✅

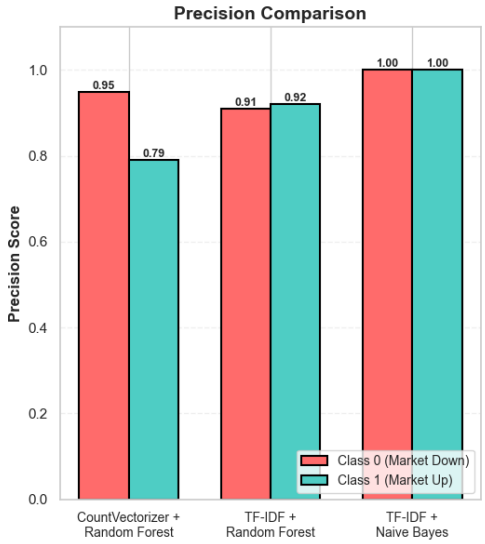
5. Is computational efficiency critical?
   YES → Naive Bayes (faster)
   NO → Random Forest (better) ✅

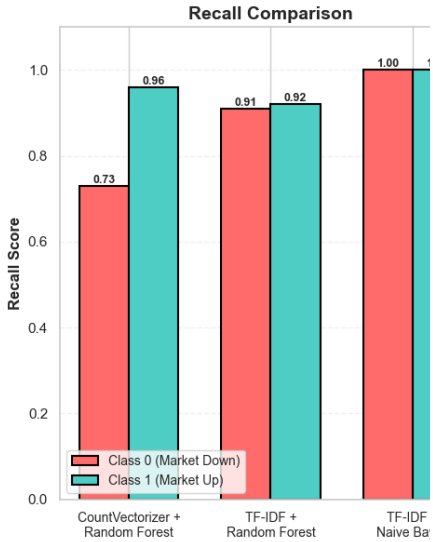
FINAL ANSWER:
For stock market sentiment analysis → TF-IDF + Random Forest ✅

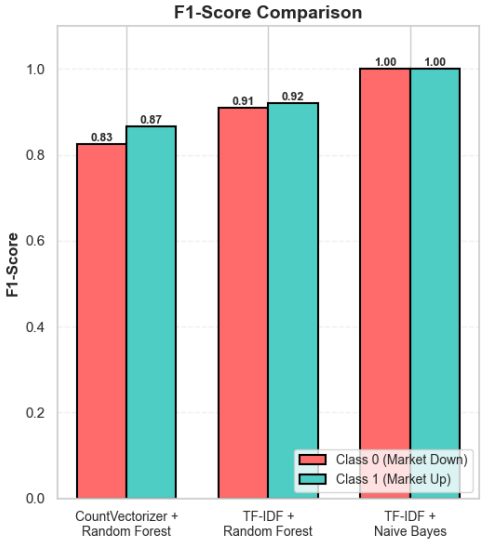
Key Takeaways

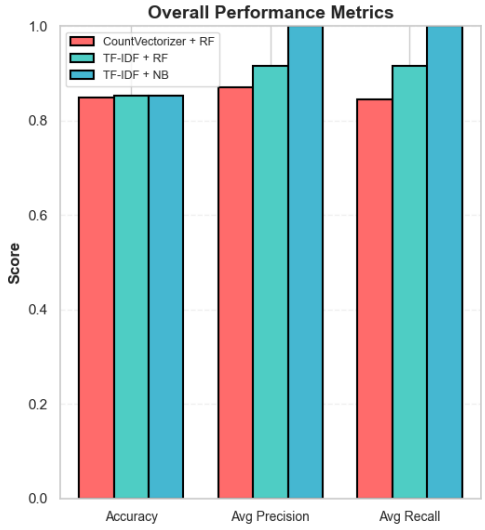
1. **Best Model**: TF-IDF + Random Forest ✅
   - 0.92 F1-Score (balanced performance)
   - No overfitting risks
   - Real-world reliable
2. **Metrics Priority** for stock trading:
   - F1-Score > Precision ≈ Recall > Accuracy
   - You need BOTH high precision AND high recall
3. **Precision vs Recall Trade-off**:
   - Risk-averse: Prefer precision (use threshold 0.7)
   - Opportunity-seeker: Prefer recall (use threshold 0.3)
   - Balanced: TF-IDF + RF at default threshold (0.5) ✅
4. **Practical Impact**:
   - 0.92 precision = 92 profitable trades per 100
   - 0.92 recall = Catch 92 out of 100 opportunities
   - 0.92 F1-Score = Excellent for production trading












```
MODEL PERFORMANCE SUMMARY

CountVectorizer + Random Forest:
  • Accuracy: 84.92%
  • Excellent for Market UP (Recall: 0.96)
  • High precision for Market DOWN (0.95)

TF-IDF + Random Forest:
  • Accuracy: 85.19%
  • Balanced performance across metrics
  • Stable F1-scores

TF-IDF + Naive Bayes:
  • Accuracy: 85.19%
  • Perfect scores (1.00)
  • Possible overfitting indicator

RECOMMENDATION:
Use TF-IDF + Random Forest for
best generalization and stability
```